

India's Energy Security – the road ahead

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India stands at a unique crossroads today. On one hand, the economy is poised to grow at more than 7% annually for a prolonged period which demands huge power consumption and on the other hand, India has to switch to cleaner sources of power generation in order to honour its commitment towards reducing carbon footprint to ultimately achieve the national goal of Net Zero emissions by 2070. Growth in GDP is directly correlated with energy consumption which increases the emission of methane (CH₄), carbon dioxide (CO₂) and other greenhouse gases (GHGs), thus accelerating global warming and climate change. Hence, a developing nation like ours (which has been a late entrant in the industrial revolution) not only has to ensure its energy security, but is also burdened with a moral obligation to strike a delicate balance between growth and emission mitigation – for the greater good of the entire world.

Global warming – a reality

When former US Vice President Al Gore produced the documentary 'An Inconvenient Truth' to highlight the dire consequences of global warming, the film's release in 2006 created quite a stir globally. In fact, it ended up winning two Academy Awards in addition to garnering multiple international accolades and recognitions. But at that time, many had dismissed it as gross fear-mongering. However, two decades down the line, the reality of climate change has started to bite. 2023 marked a milestone year which witnessed temperatures consistently exceeding the 1850-1900 pre-industrial levels. Extreme weather events have become more frequent, compelling governments worldwide to huddle up and devise strategies to combat this clear and present danger.

There is no room for debate that much of the global warming today has been a result of large-scale industrialisation and the rampant use of fossil fuels. Governments are presently faced with a unique trilemma of balancing energy security with adherence to environmentally sustainable practices while keeping in mind the affordability factor in terms of pricing power so that socio-economic disparities do not worsen.

Governments today agree that global warming is irreversible. Thus, their efforts are now directed towards slowing it down and mitigate its impact as much as possible. It is no longer enough to reduce the emissions, there is an urgent need to reverse it as well. At the COP28 climate talks in Dubai held in December 2023, more than 130 countries pledged to protect the environment and have committed to achieve the following targets by 2030:

- reduce methane emissions by minimum 30% from 2022 levels as methane is more damaging than CO₂ for its ability to deplete the ozone layer
- enable a 3X scale up of renewable energy (RE) capacity to a minimum of 11,000 GW
- double the efficiency levels of existing generating assets from 2% to 4%

Work in progress

Aligned to the 2030 targets, a lot of work is happening worldwide. The larger focus is on switching to clean energy sources. This has fast-tracked the scaling up of RE capacity. However, the intermittent nature of RE due to its dependence on natural forces remains a challenge. This is forcing both government and private sectors to opt for hybrid projects for ensuring round-the-clock (RTC) availability of power. This makes it imperative to invest in enhancement of the battery and storage capacity. Green hydrogen, as a clean energy source, is rapidly gaining traction worldwide. In parallel, bio-fuels and sustainable aviation fuel (SAF) are seen to hold a lot of promise as alternate fuels with low carbon content. Going forward, electric power will be driving the transportation sector. Blending of bio-fuels is also likely to pick up, although there exists a question mark on the sustainability of use of bio-fuel as its production competes directly with the food chain, unless the raw material is switched to agricultural waste.

Having said that, the necessary innovations or technological breakthroughs are still a few years away, and the new technologies will take time to mature and attain scale, which will ultimately pave the way for a complete switch to cleaner sources of power. Till then, a premium has to be paid for facilitating the transition because clean energy sources are yet to become cost competitive. The onus is on the governments to allocate substantial resources to make this transition happen.

Meanwhile, the systematic retirement of the old and low efficient thermal power plants which use fossil fuels, implementation of Flue-gas desulfurization (FGD) will have to be done immediately to reduce impact.

Countries' level of preparedness

Countries have agreed to reverse climate change. However, they are adopting different strategies and are on different phases in this fight, their actions mostly determined by their level of economic development. Europe has taken the lead and climate change has been on their radar for more than a decade. The governments there have been allocating substantial resources on clean energy initiatives. Those countries have also made good progress in tapping offshore wind energy, so much so that subsidy is no longer needed for it. USA has been relatively late compared to the Europe. However, the Inflation Reduction Act (IRA) introduced in 2022 has generously allocated resources to put the economy on a path towards meeting its global climate change commitments. Heavy subsidies are being offered to businesses, households, and even sub-national governments when they make investments aimed at reducing emissions and switching to clean energy. China, which produces about 30% of total global emissions (more than the emissions from the US, European Union and India combined), has pledged to reduce emissions by 65% from their 2005 level by 2030 and achieve carbon neutrality by 2060. The Middle East countries are working towards making oil & gas less carbon-intensive, besides investing heavily on solar energy, green hydrogen, blue hydrogen, blue ammonia and carbon capture projects. Biofuels, especially bioethanol, have

been a part of Brazil's national energy policies for nearly half a century. South Korea is actively pursuing a switch from RE to nuclear energy. However, African countries have made little progress on the clean energy front, except to some extent on Geo-thermal energy because of their Geological advantage. In their transition journey, the African nations will be needing active support from the developed world and the latter would need to help them for ensuring a unified fight to counter global warming.

Where does India stand?

India's energy consumption has more than doubled since 2000, driven by a growing population, industrialisation, urbanisation and rising aspiration levels. This currently stands at around 1 Gtoe (gigatonne of oil equivalent). Over 90% of India's energy needs are met by three fuels:

- Coal (46%) has underpinned the expansion of electricity generation and industry, and remains the largest single fuel in the energy mix
- Oil (24%) consumption and imports have grown rapidly on account of rising vehicle ownership and road transport use
- Biomass (21%), primarily fuelwood, makes up a declining share of the energy mix, but is still widely used as a cooking fuel. Despite recent success in expanding coverage of LPG in rural areas, 660 million Indians have not fully switched to modern, clean cooking fuels or technologies

The balance 9% requirement is met from natural gas, hydro, nuclear and RE sources.

Near universal household access to electricity was achieved in 2019. However, energy use on a per-capita basis remains well under half the global average, and there exist widespread differences in energy use and the quality of service across states and between rural and urban areas. The affordability and reliability of energy supply are key concerns for India's consumers.

Any developing economy thrives on electricity consumption. For India, while the current peak demand for electricity is around 205 GW, the National Electricity Plan for 2022-32 projects a peak demand of 277.2 GW by 2026-27 and 366.4 GW in 2031-32. India's present installed capacity stands at 442 GW within which fossil fuels account for 56% while RE sources (including hydel power) account for 42%. The Plan aims to enhance non-fossil based capacity to 57.4% by 2026-27 and to 68.4% by 2031-32. Having said that, since RE sources like solar or wind power cannot ensure stable supplies, the baseload for India's power demand (i.e. the minimum amount of power the electrical grid requires at any given time) continues to be coal-fired power plants. While internationally, many developed nations are gradually phasing out their coal-based power generation capacity, India's reliance on conventional fuels is unlikely to go away anytime soon. In comparison, about 60% of electricity generation in USA was from fossil fuel (42% from Natural Gas and 18% from Coal).

India is the 5th largest GDP (after USA, China, Germany and Japan) but India's per capita CO₂ emissions are significantly lower (2 tons) than the global average (4.66 tons). All the developed

countries emit even higher CO₂ per capita. For example, USA: 14.95 tons, China: 7.99 tons, Germany: 7.98 tons, Japan: 8.5 tons. Hence, even if the energy consumption of India rises, as India progresses towards becoming a developed nation, the per capita CO₂ emissions still be lesser than the other developed countries.

Of the two non-coal fossil fuels, namely oil & gas, oil is much more carbon-intensive, i.e. to produce the same amount of energy 33% more CO₂ is generated from oil vis-à-vis gas. Thus, natural gas, a cleaner source, will remain in demand as a transition bridging fuel and priority capital will be spent on gas production as well as for setting up the distribution infrastructure. The demand for crude will be increasingly driven by crude-to-chemicals projects, as India's per-capita consumption of petrochemical products is still much lower than that in developed nations. However, for both these non-coal fossil fuels, India is largely dependent on imports. India imports around 85% of its crude oil requirements and around 45% of its natural gas needs. This makes India particularly vulnerable to supply disruptions and price volatility in the global energy market. As a safeguard, strategic petroleum reserves (SPRs) are being built across various locations to enhance energy security in case of supply disruptions or emergencies. A more lasting solution would be to identify hydrocarbon reserves within the country. The government intends to increase the exploration acreage to 0.5 million sq.km. by 2025 and 1 million sq.km. by 2030. The 99% reduction in no-go areas or places where exploration activity was prohibited earlier has freed up 0.91 million sq.km. area for hydrocarbon exploration. The government is also setting up a National Data Repository (NDR) so that large companies can participate in energy exploration in India's sedimentary basins. Work is also underway on creating a cloud-based and artificial intelligence/machine learning (AI/ML)-powered NDR 2.0.

Constantly changing geopolitics has proved to be a stumbling block for setting up transnational gas pipelines. For India, as many as three such pipeline schemes (Myanmar-Bangladesh-India, Iran-Pakistan-India and Turkmenistan-Afghanistan-Pakistan-India) did not materialise. Now, the focus has shifted to obtain gas in LNG form, and this also allows the flexibility to change supplier sources vis-à-vis the pipeline mode where buyer gets stuck with a supplier giving the latter an undue advantage.

At the COP26 summit in Glasgow held in 2021, India made a commitment to cut its emissions to net zero by 2070. India has also set a target for itself to harness 50% of its energy requirement from RE sources and aims to cut down carbon emissions by 1 billion tonne, both by 2030.

From the present 191 GW installed capacity of non-fossil fuel based power, India is now aiming to scale it up to 500 GW by 2030. Within that, 485 GW is being envisaged to come from RE sources, up from today's capacity of 184 GW. Although this implies a more than 2.5x increase in RE capacity and thus may look a bit ambitious, these are achievable targets. India enjoys a geographical advantage that needs to be exploited. With nearly 300 sunny days every year, India's solar energy potential is huge. There is already substantial solar power generation capacity (more than 75 GW). In addition, 50 solar parks with an aggregate capacity of 37.49 GW have been approved. Also, the recently announced Pradhan Mantri Suryodaya Yojana aims

to enhance adoption of solar power usage at the family level. With a targeted installation of 10 crore solar rooftops on houses which will reduce the electricity bill of poor middle class families, this plan intends to increase the grid-connected solar rooftop capacity from the present 11 GW to 40 GW. India's installed wind energy capacity stands at 45 GW presently. With a coastline of more than 7,600 km, there is huge potential to tap offshore wind generation. An offshore target of 30 GW by 2030 has been set and the potential sites have been identified.

Clean Energy – the building blocks

Storage: The issue of intermittency of RE sources makes it imperative to explore storage solutions. Energy transition cannot happen unless the storage problem is solved. By making use of storage systems such as battery energy storage system (BESS), pumped hydro storage (PHS), hydrogen energy storage system (HESS) and compressed air energy storage (CAES), the excess RE generated during peak production can be stored for later use. This will also be the key in enhancing grid stability. However, India's perception of energy storage needs to be expanded from assets that enable RTC supply of RE to assets that can provide various grid services. Widening the use of energy storage technologies requires a favourable policy and regulatory framework, financial support, and personnel training. Storage solutions are bound to be costly initially, but over time they promise substantial returns. With increased R&D on these solutions and as economies of scale come into play, the cost of storage is expected to come down and become commercially viable. Till then, government support in terms of fiscal incentives would help.

Green hydrogen: Globally, hydrogen has emerged as a major part of the fuel mix used by industries like steel, refinery, fertilisers, etc. For its decarbonisation initiatives, India is banking majorly on green hydrogen and has announced the National Green Hydrogen Mission. India had proactively started putting in place policies such as free transmission, open access and non-applicability of cross-subsidy surcharges on RE required for producing green hydrogen. In the first phase of the mission, the focus is on replacing grey hydrogen used in oil refining and grey ammonia used in fertiliser manufacturing for domestic use. In addition, a production-linked incentive (PLI) scheme for electrolyser manufacturing has also been announced which has received an overwhelming response reflecting the growing interest in green hydrogen. Given the abundance of available land and natural resources like sunlight and wind speed, India can emerge as a global supplier of green hydrogen and green ammonia. Incidentally, India has launched the ambitious National Green Hydrogen Mission in January 2023, aiming to become a global leader in the production, utilization, and export of green hydrogen. Already, the EU member states, Japan and South Korea have evinced interest in sourcing green hydrogen and green ammonia supplies from India. According to International Energy Agency (IEA), green hydrogen and other low-carbon technologies can potentially create a USD 80 billion market in India by 2030.

Solar components: Solar energy forms a massive plank of India's transition to green energy. However, a 2.5-3x increase in solar energy capacity calls for massive import of solar modules from China. Given the growing volatility in geopolitics, India must reduce such dependency.

Presently, India has almost nil capacity in solar materials except solar modules and cells. India's capacities for manufacturing of solar wafer, polysilicon (the raw material for solar wafer production), ancillary materials are all very limited, and thus heavily dependent on imports. Even the main equipment that manufacture solar modules and upstream materials are non-existent in India, and need to be imported from Europe or China. PLI and SPECS (scheme for promotion of manufacturing of electronic components and semiconductors) are expected to play a key role towards bolstering the domestic manufacturing ecosystem for solar energy. Already, some mega projects for making Polysilicon, which is required for Solar Power Cells as well as Semiconductors have been envisaged by private developers.

Nuclear Energy: Nuclear energy is also part of India's energy mix, although its share is not even 2% of the total. While this is viewed as best option for clean energy, presently public sector dominates this sector. India has private companies involved in construction and manufacturing of machine parts for nuclear power plants, but no private company owns and operates a nuclear power plant presently. The world is witnessing a growing trend of small capacity atomic power plants, popularly called Small Modular Reactors (SMRs), with their unique features of modularity, scalability, small footprint and improved safety for repurposing of retiring coal-based thermal power stations. India can encourage private sector participation for development of SMRs. However, uranium - the raw material - is in short supply in India and we need to depend on import of natural uranium from France, Canada, Russia, Kazakhstan and Uzbekistan. This calls for large scale mineral prospecting within India. Another alternative is the Nuclear fusion technology, which does not depend on nuclear fuel. However, making the fusion technology viable at a commercial scale will take some more time. Further, funding on the research and development of Nuclear fusion technology may speed up the commercial deployment.

Energy efficiency: Improving energy efficiency across sectors like industry, transportation and buildings is a crucial part of India's energy policy. Initiatives such as the Perform, Achieve and Trade (PAT) scheme have been implemented to promote energy efficiency in industries. While India continues to rely on coal for a significant portion of its energy needs, there are efforts to increase the efficiency of coal-based power plants and explore cleaner technologies such as carbon capture and storage (CCS).

The missing link

In order to ensure that India does not become energy deficient, Nation Electricity Plan targets the following additional generation capacity by 2032:

- Coal / Lignite	:	49 GW
- Nuclear	:	13 GW
- Hydro	:	16 GW
- Solar	:	310 GW
- Wind	:	82 GW

As would be noted, the lion's share is coming from RE sources.

While there is clarity on the way forward, the Indian government would need to address a couple of issues to make this green transition a reality:

- adopting advanced technologies and innovations prevalent in developed nations to enhance energy efficiency, reduce emissions, and optimise resource utilisation calls for substantial investments in R&D, skill development and technology transfer
- transition to clean and green energy has to be made affordable for the masses

Everything will ultimately boil down to how the finances needed for this transition get mobilised. As per IEA projections, India will be needing an average annual investment of USD 160 billion across its energy economy between now and 2030 to be on target for its ultimate net zero emission goal by 2070. This is a massive amount keeping in mind that the budget outlay for the entire infrastructure sector for 2024-25 stood at around USD 135 billion. This can be immensely challenging when access to low cost global funds is virtually over. Indian government would need to leverage its diplomatic goodwill to scout for and tie up long term funds. In addition, government would need to incentivise the private sector (both domestic and foreign) to play an active role for this transition to play out successfully – be it for collaboration on R&D, skilling, investing in setting up green-tech manufacturing units, and even financing. To make mass adoption of clean energy technology a reality, our policy-makers should ensure availability of soft loans and other fiscal incentives for both manufacturers and users of such technology and also allow accelerated depreciation of such assets. To counter climate change at the international level, some progress has already been made in terms of setting up early warning systems, building climate resilient infrastructure, and work on ecosystem restoration. But much more needs to be done if we are serious about leaving a liveable planet to our next generations. India, by virtue of its growing stature as an economic force, enjoys clout and can influence global policy makers. This was evident in the way the Indian government had taken the lead in setting up the International Solar Alliance in 2015. Now is the time for India to appeal to all countries to jointly put in place an international institutional mechanism that can bring together governments, corporate sector and the top scientific brains to seamlessly collaborate on technology R&D towards clean and green energy and cost optimisation. In addition, India should also propose the setting up of dedicated multinational vehicles that can finance the laggards for playing catch-up.